The Importance of Data Modelling in MongoDB

MongoDB achieves scalability with ease due to its unique architecture. The key component of this architecture is the data model, which is based on schema-less documents and collections. While adopting MongoDB in any application, it is important to base it on data Modelling.

MongoDB is one of the most popular NoSQL databases around and is gaining popularity exponentially. Many more developers have started using it and increasing numbers of applications are built using MongoDB. As organisations adopt MongoDB as an integral architectural component of their solutions, it is important that they build it on the solid foundation of data Modelling.

Data Modelling is the first step in using any database, be it relational or NoSQL. It refers to the process of creating database design iteratively to meet the application's needs. It involves analysis and depiction of data entities and their relationships for an application.

Traditional databases are primarily relational in nature, where the database schema is defined at design time based on data objects. The data structure is static in nature, and data should follow the rules to be stored and retrieved using this static schema. If data does not follow the database schema, it cannot be stored in the database. Moreover, database structures are normalised and decomposed into multiple smaller tables to avoid data repetitions and implement the get-what-you-want pattern of data retrieval. These tables have relationships built into them to enforce their consistency, integrity, atomicity and durability.

On the other hand, NoSQL databases are generally schema-less. Although there is a database schema defined to start with, data does not have to follow the schema strictly. NoSQL databases accept data of any structure within their collections (tables in the relational world), irrespective of whether it matches their schema or not. In other words, they are flexible regarding their enforcement of schema to data. They do not have inbuilt inherent relationships between their collections; rather, applications need to implement the necessary logic for atomicity and consistency of data between collections. And there are no inbuilt constructs to join data from multiple collections.

MongoDB provides two different ways to create relationships between data objects – References and Embedment.

Data Modelling in the NoSQL world

Data Modelling is equally important in the NoSQL world as it is in the relational world. There are important facets of applications that cannot be realised without implementing proper and optimised data models. Data Modelling is not an afterthought. It is an important process in the application planning and design phases. Some of the important reasons